

Process Characterization Report

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCR-003

Process Development / Manufacturing Science & Technology (MSAT)

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Executive summary

This report documents the Stage 1 (Process Design) characterization of the **A-Mab production bioreactor (Step 3)**, executed under **PCP-003**. A 19-run screening design and a 28-run response-surface design on a qualified 2 L scale-down model related the culture parameters to the glycosylation, charge-variant and aggregate CQAs and established the upstream design space.

Key outcomes

- Culture pH, temperature, dissolved CO₂, osmolality and culture duration are classified **well-controlled CPP (WC-CPP)** — they significantly affect the glycan and charge-variant CQAs but are reliably controlled within the design space. Dissolved oxygen, initial viable cell concentration and nutrient feed-1 volume are **KPP**; basal medium concentration is **GPP**.
- The nominal fed-batch reaches a peak viable cell density of **13.48 ×10⁶ cells/mL** and a harvest titer of **4.37 g/L** at **67.1%** final viability.
- All in-scope CQAs meet their acceptance criteria at set-point and across the normal operating range; the minimum commercial-scale process capability across the bioreactor-set CQAs is **Cpk = 1.94**.
- A multivariate design space in culture pH, temperature, duration and CO₂ was established over which every cell-culture CQA is satisfied simultaneously.

The production bioreactor is well understood; the design space, classifications and proven acceptable ranges support the Stage 2 operating ranges and the overall control strategy.

1 Related documents

Table 2: Related documents in the A-Mab process-characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-003	Process Characterization Plan (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Purpose and scope

This report presents the results of the production-bioreactor characterization defined in **PCP-003**, following PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023) and the A-Mab methodology (CMC

Biotech Working Group 2009). Scope, quality attributes and parameter ranges are as defined in PCP-003 and are not restated except where needed to interpret the results.

3 Study execution

Characterization was performed on a 2 L scale-down model qualified against at-scale reference data for viable cell density, viability, titer and the input and output quality attributes, confirming the model is representative and suitable to support design-space and PAR claims (Parenteral Drug Association 2013). Two designs were executed on the SDM: a resolution-V two-level fractional factorial screening design (19 runs in culture pH, temperature, dissolved CO₂, osmolality and culture duration) and a face-centred central-composite response-surface design (28 runs in the four dominant factors). The response-surface models are adequate across the CQAs (Table 3).

Table 3: Response-surface model adequacy (coefficient of determination) for the bioreactor CQAs.

Response	Model R ²
Afucosylation	0.965
Galactosylation	0.998
High mannose	0.942
Acidic variants	0.994
Aggregates (HMW)	0.995

4 Parameters and classification

Table 4 lists the parameters studied with their set-points, normal operating ranges, proven acceptable ranges and final classification.

Table 4: Production-bioreactor parameters, ranges and post-characterization classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO ₂ (pCO ₂)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	multivariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	multivariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	multivariate

5 Nominal batch performance

The nominal fed-batch profile (Figure 1) reaches a peak viable cell density of 13.48×10^6 cells/mL and a harvest titer of 4.37 g/L at 67.1% final viability at the nominal 17-day harvest.

Production bioreactor — nominal fed-batch profile (set-point operation)

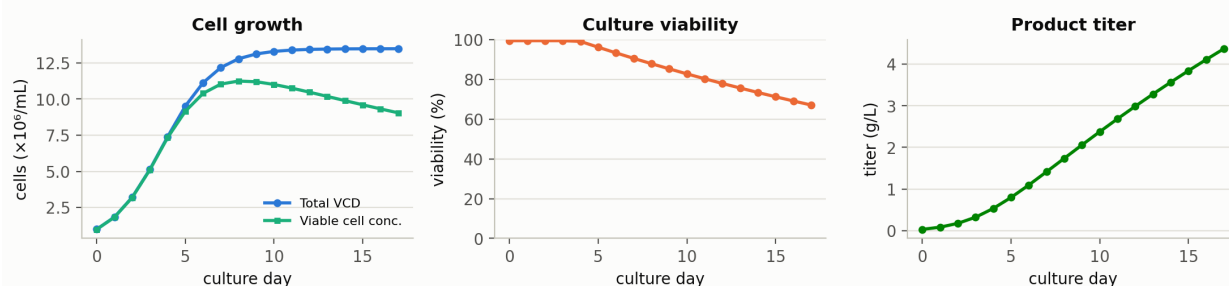


Figure 1: Nominal production-bioreactor time course: viable cell density, viability and titer.

6 Screening results

Screening identified culture pH, temperature, dissolved CO₂, osmolality and culture duration as significant for the glycan and charge-variant CQAs (Figure 2). The largest standardized effects on afucosylation and galactosylation are given in Table 5 and Table 6.

Bioreactor screening DoE — standardized effects (red = significant, $p < 0.05$)

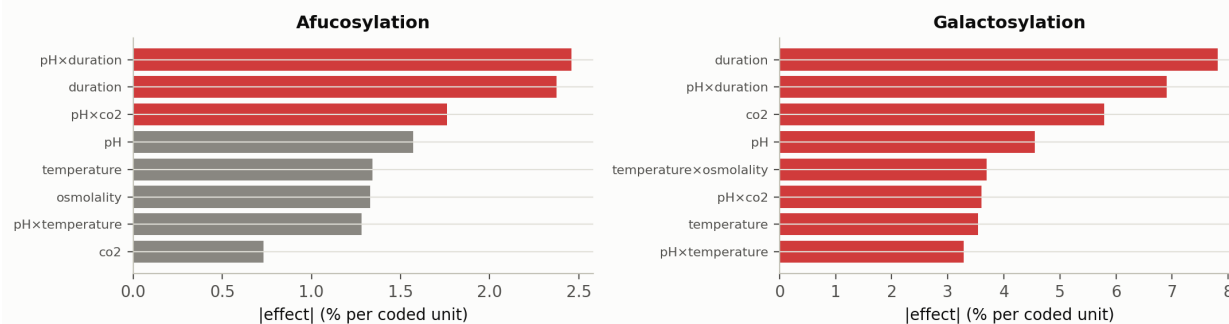


Figure 2: Standardized effects (screening DoE) for afucosylation and galactosylation; red bars are significant ($p < 0.05$).

Table 5: Largest standardized effects on afucosylation (bioreactor screening DoE).

Term	Effect	p-value	signif.
pH $\times$ duration	2.46	0.0191	yes
duration	-2.37	0.0209	yes
pH $\times$ co2	-1.76	0.0455	yes
pH	-1.57	0.06	
temperature	-1.34	0.0858	
osmolality	-1.33	0.0878	

Table 6: Largest standardized effects on galactosylation (bioreactor screening DoE).

Term	Effect	p-value	signif.
duration	-7.82	0.000264	yes
pH $\times$ duration	6.9	0.000382	yes
co2	-5.79	0.000644	yes
pH	-4.56	0.00131	yes
temperature $\times$ osmolality	3.7	0.00242	yes
pH $\times$ co2	3.6	0.00261	yes

Lower pH and lower temperature increase specific productivity, culture duration and titer, with a modest increase in afucosylation and a decrease in galactosylation; extended culture duration lowers afucosylation and galactosylation and raises the harvest HCP and DNA load as viability declines.

7 Response surface and design space

The response-surface design quantified the main effects, two-factor interactions and curvature and defined the multivariate design space (Figure 3). Within the characterized ranges all cell-culture CQAs remain within acceptance simultaneously, defining a robust design space in culture pH, temperature, duration and dissolved CO ; the shaded region satisfies every CQA at once.

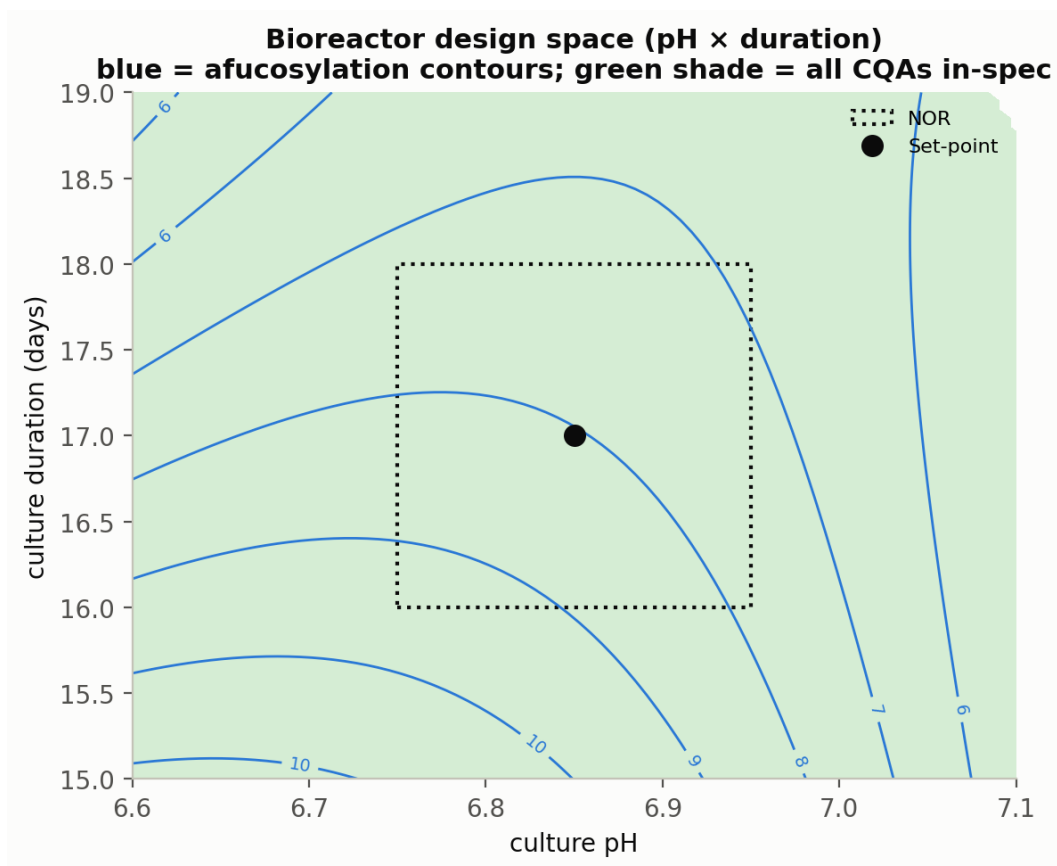


Figure 3: Production-bioreactor design space (pH × culture duration); the shaded region satisfies all cell-culture CQAs simultaneously. Blue contours are afucosylation (%).

8 CQA outcomes and process capability

A Monte-Carlo simulation of 2,000 commercial-scale batches, with every parameter varied within its normal operating range, was used to estimate the bioreactor-set CQA distributions and process capability (Table 7). Capability is evaluated one-sided for impurities (upper limit) and two-sided for target-range attributes.

Table 7: Commercial-scale process capability for the bioreactor-set CQAs.

CQA	Crit.	Spec	Acceptance	Mean $\pm$ SD	Cpk
Afucosylation	H	two_sided	2–13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10–40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3–10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.11
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14

All in-scope CQAs meet their acceptance criteria, with a minimum capability of $C_{pk} = 1.94$ across the bioreactor-set CQAs (high mannose is the tightest, being a Very-High-criticality attribute controlled to a narrow range).

9 Parameter classification and rationale

Culture pH, temperature, dissolved CO₂, osmolality and culture duration significantly affect the glycan and charge-variant CQAs but are reliably held within the design space by closed-loop control and in-process monitoring; they are therefore **WC-CPP**. Dissolved oxygen, initial viable cell concentration and nutrient feed-1 volume affect peak viable cell density, titer and process consistency without a significant CQA impact and are **KPP**. Basal medium concentration showed no meaningful impact over a wide range and is **GPP**.

10 Conclusions

The production bioreactor is well understood and robust: the relationships between the culture parameters and the glycan, charge-variant and aggregate CQAs are quantified, a multivariate design space is established, and every parameter range is justified by data. The design space, classifications and proven acceptable ranges provide the basis for the Stage 2 operating ranges and feed the overall control strategy. This report rolls up into the Process Characterization Master Report (**PCMR-001**) and provides the parameter risk basis for the post-characterization process risk assessment.

Reproducibility

Every figure, table and number in this report is generated from the seeded A-Mab process model (master seed 20240724); the datasets and figures are produced by `make all` in the repository and are shared with the consolidated process-characterization report.

References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS. International Society for Pharmaceutical Engineering. 2023. “Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation.” ISPE.
Parenteral Drug Association. 2013. “Technical Report No. 60: Process Validation — a Lifecycle Approach.” PDA.